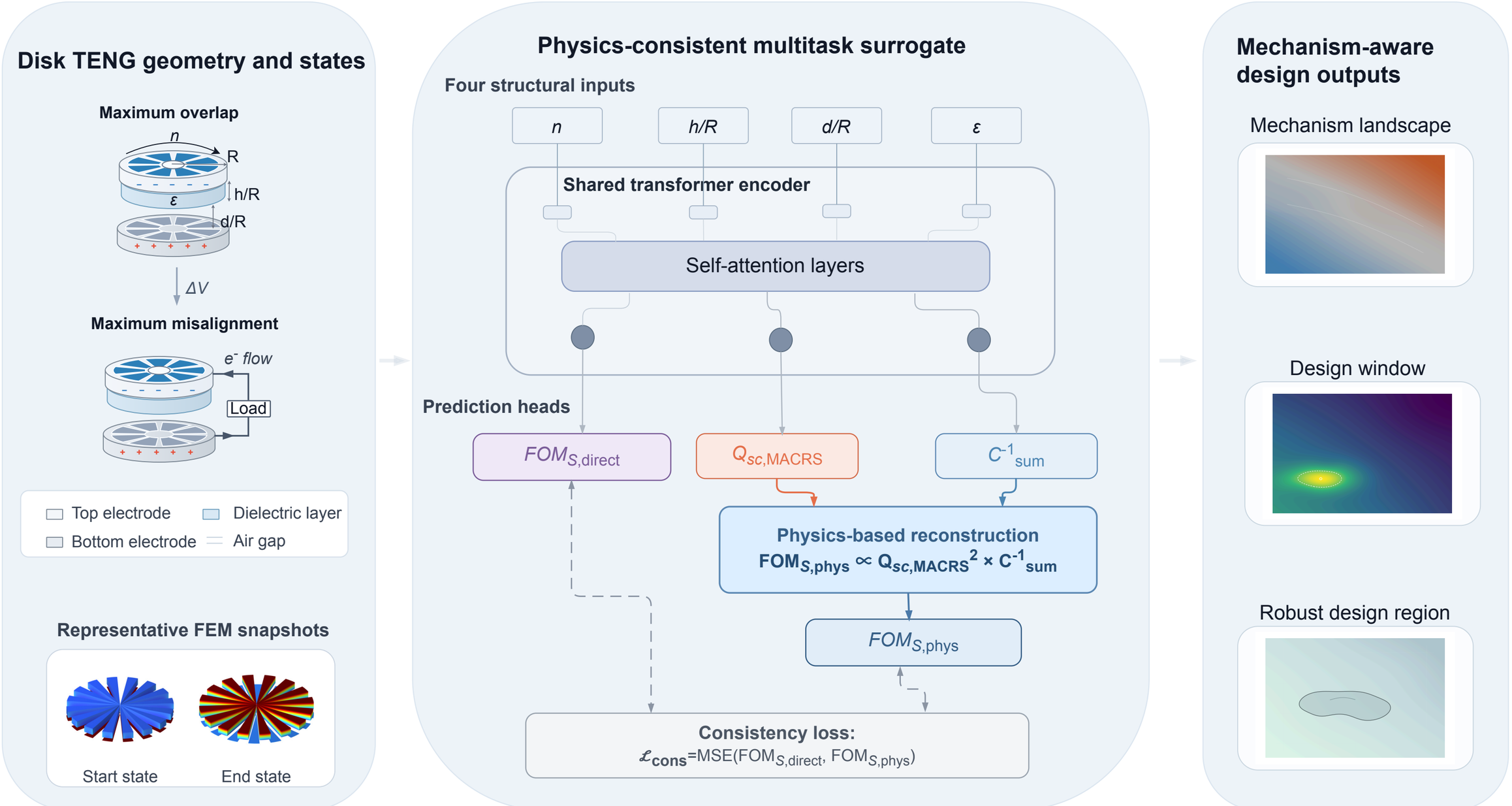
Disk TENG geometry and states
Maximum overlap
n
R
h/R
ε
d/R
ΔV
Maximum misalignment
e⁻ flow
Load
Top electrode
Dielectric layer
Bottom electrode
Air gap
Representative FEM snapshots
Start state
End state
Physics-consistent multitask surrogate
Four structural inputs
n
h/R
d/R
ε
Shared transformer encoder
Self-attention layers
Prediction heads
FOM_S,direct
Q_sc,MACRS
C⁻¹_sum
Physics-based reconstruction
FOM_S,phys ∝ Q_sc,MACRS² × C⁻¹_sum
FOM_S,phys
Consistency loss:
ℒ_cons=MSE(FOM_S,direct, FOM_S,phys)
Mechanism-aware design outputs
Mechanism landscape
Design window
Robust design region